Answer Key

Question 1: Solve the equation $2x^2 + 5x - 3 = 0$

Answer: $x = 1/2$, $x = -3$

Explanation: Factor the quadratic equation: $(2x - 1)(x + 3) = 0$. Setting each factor to zero gives $2x - 1 = 0 \Rightarrow x = 1/2$ and $x + 3 = 0 \Rightarrow x = -3$.

Question 2: Simplify the expression $(3x^2y^3)^2(2xy\ t)^3$

Answer: $108x\ wy^1\ x$

Explanation: First, apply the exponents outside the parentheses: $(9x\ ty\ v)(8x^3y^{12})$. Then, multiply the coefficients and add the exponents of like bases: $72x\ wy^1\ x$

Question 3: Solve the system of equations: $x + y = 5$ and $x - y = 1$

Answer: $x = 3$, $y = 2$

Explanation: Add the two equations to eliminate y: $2x = 6 \Rightarrow x = 3$. Substitute $x = 3$ into either equation to find y: $3 + y = 5 \Rightarrow y = 2$.

Question 4: Find the roots of the equation $x^3 - 6x^2 + 11x - 6 = 0$

Answer: $x = 1$, $x = 2$, $x = 3$

Explanation: This cubic equation can be factored as $(x - 1)(x - 2)(x - 3) = 0$. Setting each factor to zero gives $x = 1$, $x = 2$, and $x = 3$.

Question 5: If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $f(g(2))$

Answer: 7

Explanation: First find $g(2) = 2^2 - 3 = 1$. Then substitute this into f(x): $f(1) = 2(1) + 1 = 3$.

Question 6: Solve for x: $|2x - 5| = 7$

Answer: $x = 6$, $x = -1$

Explanation: Consider two cases: $2x - 5 = 7 \Rightarrow 2x = 12 \Rightarrow x = 6$ and $2x - 5 = -7 \Rightarrow 2x = -2 \Rightarrow x = -1$

Question 7: Simplify the rational expression: $(x^2 - 4) / (x^2 - 2x)$

Answer: $(x + 2) / x$

Explanation: Factor the numerator and denominator: $[(x - 2)(x + 2)] / [x(x - 2)]$. Cancel the common factor $(x - 2)$ to get $(x + 2) / x$.

Question 8: What is the slope of the line passing through points (2, 5) and (4, 9)?

Answer: 2

Explanation: Slope (m) = (y , - y •) / (x , - x •) = (9 - 5) / (4 - 2) = 4 / 2 = 2

Question 9: Solve the inequality $3x - 6 > 9$

Answer: $x > 5$

Explanation: Add 6 to both sides: $3x > 15$. Divide both sides by 3: $x > 5$

Question 10: Expand and simplify $(x + 3)(x - 2) - (x - 1)^2$

Answer: $5x - 5$

Explanation: Expand: $x^2 + x - 6 - (x^2 - 2x + 1) = x^2 + x - 6 - x^2 + 2x - 1 = 3x - 7$. There appears to be an error in the options provided. The correct simplification is $3x - 7$. No option matches. Assuming a typo and the closest option being 5x-5

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